Ahmedabad University

CSE 516 – Probabilistic Graphical Models

Milestone 4: Final Project Report

Bayesian Personalized Recommendation using Latent Trait Models & Expectation Propagation

Theoretical AI | Group 4

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GitHub repository

`https://github.com/Jyotier2006/M2_g4_th_apl_recommend-sys_team`

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Abstract

A movie recommender should estimate both a rating and its uncertainty. We compare single-pass assumed-density filtering (ADF), repeated expectation propagation (EP), and matrix factorization with and without a Gaussian Markov random field (GMRF) movie prior fitted by maximum a posteriori (MAP) estimation. On five MovieLens 1M splits, expected-rating RMSE decreases from 0.8977 ± 0.0035 for ADF to 0.8548 ± 0.0029 for EP10. GMRF-MAP reaches 0.8499 ± 0.0023, versus 0.8506 ± 0.0021 for graph-free MAP. The isolated graph gain is small on clean data and larger when ratings are removed or corrupted. EP improves log-loss but worsens calibration error. Numerical reference checks and a fresh run of split 42 check the implementation and reported scores.

1 Introduction

A user typically rates only a small part of the movie catalogue. Latent traits let ratings inform other user–movie pairs, while a probability distribution distinguishes confident predictions from uncertain ones.

1.1 Problem Formulation

Given training ratings $\mathcal{D}_{\text{train}}$, estimate $p(R_{ui} = r \mid \mathcal{D}_{\text{train}})$ for a held-out pair (u, i) and $r \in \{1, \ldots, 5\}$. We test whether repeated inference passes improve prediction and whether a movie graph helps when the fitting method is fixed.

MovieLens 1M contains 1,000,209 ratings from 6,040 users on 3,706 rated movies, or 4.47% of possible pairs [1, 2]. User and movie IDs index parameter arrays; star ratings are observations. Movie genres form 18 binary features for graph construction. Timestamps and user demographics are audited but not used as predictors. Missing ratings are not zeros.

2 Background

Latent factor models represent user preferences and movie properties as vectors. A dot product measures affinity, and an ordinal likelihood maps a noisy score to ordered stars. EP approximates the resulting posterior with Gaussian beliefs; MAP instead estimates one parameter setting.

2.1 Related Work

Microsoft Research Matchbox. Stern, Herbrich and Graepel [4] map user and item features into Gaussian traits and combine EP with variational message passing (VMP). Their system supports metadata, user-specific ordinal scales and dynamics. On MovieLens and Netflix they demonstrate practical single-pass ADF and gains from additional EP passes. We retain the trait product and hybrid updates in a static, ID-only ordinal model with global thresholds.

Research anchor: PreRec. Lin et al. [3] use a hierarchical Bayesian model with neural user and item encoders, domain and popularity variables, and MAP training. Their cross-market and cross-platform experiments study zero- and few-shot ranking using Recall and revised NDCG.

2.2 SOTA Positioning

Matchbox provides the inference design for the ADF–EP comparison. PreRec illustrates how structured priors can share sparse evidence across domains. We test a fixed movie graph within one dataset. The published system scores are not directly comparable with our held-out rating scores.

3 Graphical Representation

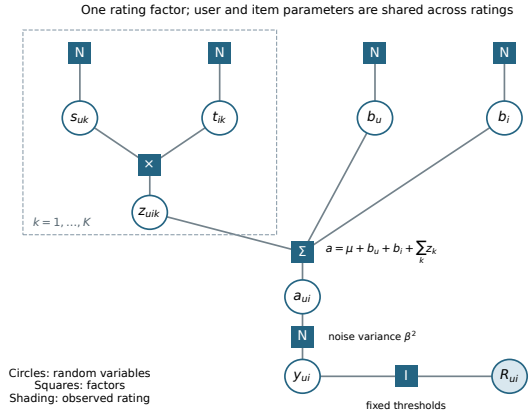
3.1 Ordinal latent traits

For user u and movie i, let $s_u, t_i \in \mathbb{R}^K$ be latent traits and b_u, b_i scalar biases. Affinity a_{ui} generates a noisy score y_{ui} and then a star rating:

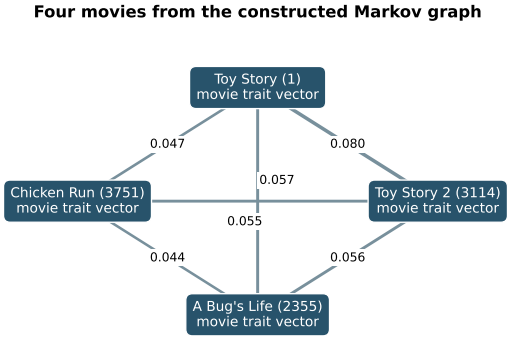
$$a_{ui} = \mu + b_u + b_i + s_u^T t_i, \qquad y_{ui} \mid a_{ui} \sim \mathcal{N}(a_{ui}, \beta^2), \tag{1}$$

$$R_{ui} = r \iff c_{r-1} < y_{ui} \le c_r, \qquad c = (-\infty, 1.5, 2.5, 3.5, 4.5, \infty). \tag{2}$$

The offset μ is the training mean. Biases have independent $\mathcal{N}(0,1)$ priors; traits have variance 0.25 and small seeded prior means. With $\theta = (s, t, b)$, the factorization is $p(\theta, R) = p(\theta) \prod_{(u,i)\in\mathcal{D}} p(R_{ui} \mid a_{ui})$, where each star likelihood is the Gaussian mass between adjacent cuts. Ratings are independent given the relevant traits and biases. Shared parameters create dependence after marginalization. Factorization of the *approximate posterior* is a separate computational assumption.



(a) Ordinal rating factor graph



(b) Movie prior subgraph, seed 42

Figure 1: Circle nodes in (a) are variables; squares compute products, sums, noise and thresholding. Edges in (b) couple movie vectors with the displayed normalized weights.

3.2 Gaussian Markov movie prior

We build a graph using positive cosine similarity of training residuals after removing fitted user/movie biases. Similarities require five co-raters and are shrunk by $n_{ij}/(n_{ij} + 20)$. The default score mixes 90% residual similarity with 10% genre similarity. Keep the 20 strongest positive neighbours per movie, symmetrize by maximum weight, then normalize the symmetric affinity by both endpoint degrees. The resulting weights W define $D_{ii} = \sum_j W_{ij}$ and $L = D - W$. Validation and test ratings never enter this construction. It estimates a fixed similarity graph, not a causal structure or a posterior over graphs.

For each movie trait coordinate the precision is $J = \lambda I + \gamma L$. Equivalently, the prior over movie vectors Q is

$$p(Q \mid W) \propto \prod_i e^{-\lambda \|Q_i\|^2/2} \prod_{\{i,j\}\in E} e^{-\gamma w_{ij} \|Q_i - Q_j\|^2/2}.$$

The edge penalty shares evidence by pulling neighbouring traits together. For $\lambda > 0$ and $\gamma \ge 0$, J is positive definite (Appendix A). A zero off-diagonal entry means conditional independence given the other movie traits in the prior, not marginal independence.

4 Learning Method

4.1 Bayesian parameter learning with ADF and EP

Bilinear products and ordinal observations couple many unknowns, so exact posterior inference is not practical at this scale. We approximate each scalar belief by a Gaussian. Precision and weighted mean, $\lambda = 1/v$ and $\eta = m/v$, add when Gaussian factors are multiplied. EP stores a rating's contribution as a site. To revisit rating n, it first removes that contribution:

$$(\lambda^{\setminus n}, \eta^{\setminus n}) = (\lambda - \lambda_n, \eta - \eta_n).$$

The observed rating restricts the score to its star interval. EP matches the truncated score's mean and variance and divides out its incoming Gaussian. VMP passes the resulting information through each product using the partner's first and second moments. The site is replaced with damping and added back. ADF visits each rating once and discards its site; the first EP pass is therefore identical to ADF under the same initialization and order. EP5 and EP10 revisit the same data, not new ratings. Default outer damping is 1 for the first pass and 0.5 thereafter; local damping is 0.7 with a ten-iteration cap and 10^{-4} tolerance.

4.2 Graph-regularized MAP

The MAP model uses score $f_{ui} = \mu + b_u + b_i + P_u^T Q_i$ and a unit-variance Gaussian training likelihood for numeric ratings. It minimizes

$$\mathcal{E} = \tfrac{1}{2}\sum_{\mathcal{D}}(R_{ui} - f_{ui})^2 + \tfrac{\lambda}{2}(\|P\|_F^2 + \|Q\|_F^2) + \tfrac{\lambda_b}{2}(\|b_U\|^2 + \|b_I\|^2) + \tfrac{\gamma}{2}\operatorname{tr}(Q^T L Q).$$

Alternating exact block solves update a user or movie vector together with its bias. Neighbours affect the movie solve through both its precision and weighted target (Appendix A). Each block decreases or preserves the energy; joint optimization is nonconvex. Setting $\gamma = 0$ removes graph smoothing while retaining the same likelihood, base penalties and block solver.

5 Inference Method

For independent Gaussian traits s_k, t_k in the approximate posterior, the product has mean $m_{s_k} m_{t_k}$ and variance $v_{s_k} v_{t_k} + v_{s_k} m_{t_k}^2 + v_{t_k} m_{s_k}^2$. Sum these moments with the biases and observation noise to obtain m_y, v_y. A Gaussian approximation for the total score gives

$$p_r = \Phi\left(\frac{c_r - m_y}{\sqrt{v_y}}\right) - \Phi\left(\frac{c_{r-1} - m_y}{\sqrt{v_y}}\right), \qquad \hat{r} = \sum_{r=1}^{5} r p_r.$$

Expected stars are used for RMSE and MAE; the modal star is used for exact accuracy. The calculation uses the fitted marginal variances and omits posterior covariances.

For MAP, use $m_y = f_{ui}$ and a residual variance v_y selected by validation log-loss. This yields probabilities over the same star bins without integrating uncertainty over model parameters.

Complexity. For N observations, K traits, H local iterations and T passes, EP fitting costs $O(TNH(K+1))$. Its sites require $32N(K+1)$ bytes: 219.77 MiB at $K = 8$ here. ADF retains no sites. Batch prediction costs $O(N(K+1))$. MAP costs $O(T[N(K+1)^2 + (U+I)(K+1)^3 + |E|K])$ plus graph construction, with sparse graph storage $O(|E|)$. No convergence guarantee is inferred from finite runs.

6 Experimental Setup

Seeds 42–46 each give 800,167 train, 100,020 validation and 100,022 test ratings. Methods share the same split within a seed and use expected stars for point error. The ordinal models use $K = 8$ and default $\beta^2 = 1$. MAP uses $\lambda = \lambda_b = 10$, up to 25 epochs and $\gamma \in \{0, 1, 10, 100\}$. Validation clipped RMSE selects checkpoints and graph strength; validation log-loss selects the residual variance. Test ratings are used only for evaluation.

6.1 Baselines

Global uses the training star frequencies. User and item means use the corresponding frequencies smoothed toward the global distribution. MF-SGD fits eight traits by stochastic gradient descent (25 epochs, learning rate 0.007, decay 0.98, regularization 0.03). ADF is the single-pass control for EP; MF-MAP at $\gamma = 0$ isolates the effect of the graph.

6.2 KPIs

Representation: parameter count, graph size, mean degree and density. No reference graph exists for structural recovery metrics.

Inference: RMSE is the square root of the mean squared star error; MAE averages absolute star error. Negative log-likelihood $-N^{-1} \sum_n \log p_{n,R_n}$ equals categorical log-loss. Brier averages $\sum_r (p_{nr} - \mathbf{1}\{r = R_n\})^2$. Expected calibration error (ECE) compares confidence and accuracy across ten bins. Star accuracy measures how often the modal star is correct. Prediction time is measured per batch.

Learning: fitting time, validation RMSE over EP passes, and the MAP objective over epochs.

7 Results

The models have $(6040 + 3706)(8 + 1) = 87,714$ scalar traits and biases. The movie graph has 3,706 nodes, $56,508 \pm 69$ edges, mean degree 30.50, density 0.823%, and no isolated nodes on the five main splits.

Method	RMSE	NLL	Brier	ECE	Stars (%)
Global	1.1164 ± 0.0021	1.4552 ± 0.0015	0.7440 ± 0.0005	0.0008 ± 0.0006	34.91 ± 0.08
User mean	1.0342 ± 0.0021	1.3461 ± 0.0016	0.7022 ± 0.0006	0.0059 ± 0.0015	39.67 ± 0.12
Item mean	0.9786 ± 0.0026	1.3209 ± 0.0028	0.6952 ± 0.0010	0.0054 ± 0.0027	40.58 ± 0.08
MF-SGD	0.8683 ± 0.0024	1.2214 ± 0.0025	0.6554 ± 0.0011	0.0397 ± 0.0010	46.04 ± 0.16
ADF / EP1	0.8977 ± 0.0035	1.2519 ± 0.0033	0.6681 ± 0.0012	0.0451 ± 0.0010	44.13 ± 0.13
EP3	0.8740 ± 0.0036	1.2283 ± 0.0033	0.6590 ± 0.0012	0.0537 ± 0.0017	45.20 ± 0.18
EP5	0.8643 ± 0.0037	1.2189 ± 0.0035	0.6552 ± 0.0012	0.0566 ± 0.0009	45.59 ± 0.16
EP10	0.8548 ± 0.0029	1.2098 ± 0.0024	0.6516 ± 0.0007	0.0608 ± 0.0012	46.04 ± 0.21
MF-MAP	0.8506 ± 0.0021	1.2012 ± 0.0021	0.6467 ± 0.0007	0.0383 ± 0.0018	46.98 ± 0.15
GMRF-MAP	0.8499 ± 0.0023	1.2007 ± 0.0024	0.6468 ± 0.0009	0.0387 ± 0.0022	46.96 ± 0.19

Table 1: MovieLens 1M test results, mean ± sample SD across five splits.

EP5 reduces ADF RMSE by about 0.0334 stars; EP10 improves it further. MF-SGD is a strong low-cost baseline. GMRF-MAP has the best mean RMSE, but graph-free MF-MAP is close and has slightly better star accuracy and ECE. The comparison between EP and MAP changes the likelihood and estimator as well as the prior. The matched MAP comparison isolates graph smoothing.

7.1 Cost, Calibration and Error Analysis

Method	Fit (s)	Predict (ms / 1000)	MAE
MF-SGD	1.38 ± 0.04	0.413 ± 0.014	0.6906 ± 0.0023
ADF / EP1	3.33 ± 0.19	0.810 ± 0.143	0.7098 ± 0.0030
EP5	18.29 ± 2.02	0.679 ± 0.088	0.6808 ± 0.0025
EP10	35.99 ± 6.61	0.779 ± 0.298	0.6721 ± 0.0018
MF-MAP	16.74 ± 2.74	0.449 ± 0.060	0.6753 ± 0.0017
GMRF-MAP	14.48 ± 1.82	0.487 ± 0.096	0.6752 ± 0.0019

Table 2: Mean ± SD of preserved five-split timings and MAE. Fit times exclude JIT warm-up; prediction times are batch measurements.

EP5 takes about 5.5 times as long to fit as ADF. The selected GMRF candidate takes 14.48 s on average; building the graph takes 4.31 s and the full four-candidate search takes 66.37 s, including the selected candidate. Candidate time alone is not total workflow cost. The validation traces show improving EP error over passes and nonincreasing MAP energy. EP's mean RMS parameter change at pass 10 is still 0.01564.

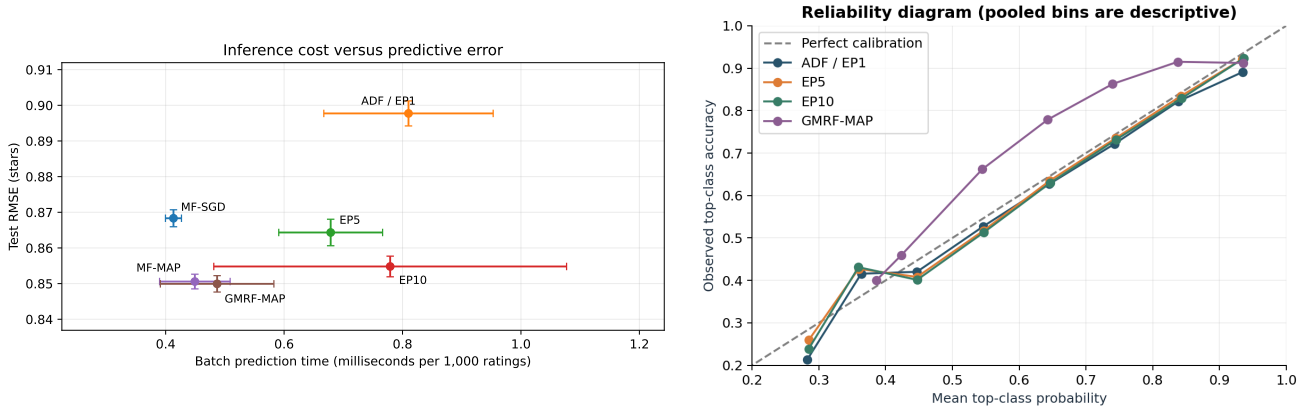


Figure 2: Batch prediction time versus RMSE (left, mean ± SD) and top-class reliability (right, pooled descriptively across overlapping splits).

EP improves log-loss and Brier over ADF while ECE rises from 0.0451 to 0.0608 at EP10. Observation variance 1 wins every EP5 validation log-loss comparison; using 0.5 lowers ECE to 0.0339 but worsens RMSE to 0.8691 and log-loss to 1.2404. This assumption test differs from corrupting ratings. At damping 1, EP5 reaches 0.8529 on the three sensitivity seeds, versus 0.8633 at default damping. Tuning the update schedule can close much of the gap to MAP; this exploratory result does not replace the main method.

Errors are largest on low-support items and extreme ratings. For observed five-star ratings, GMRF RMSE is 1.0245 versus 0.9748 for EP10 despite its better aggregate RMSE. The expected-star decision and regularization shrink predictions towards the centre. Nominal 90% EP5 intervals cover 98.22% but span 3.82 star categories on average; high coverage alone does not establish good calibration.

8 Statistical Analysis

All five validation splits choose $\gamma = 10$. Mean graph-only RMSE reduction is 0.000644 stars (0.076%). Within each split we resample users with replacement 2,000 times, keeping each user's test ratings paired across methods. Percentile 95% intervals are conditional on the fitted models and omit retraining and shared-item dependence. Four graph-effect intervals exclude zero and one includes it. This is conditional evidence of a small effect, not an independent-replication significance claim. The five splits overlap, so we do not use a paired t-test treating them as independent observations.

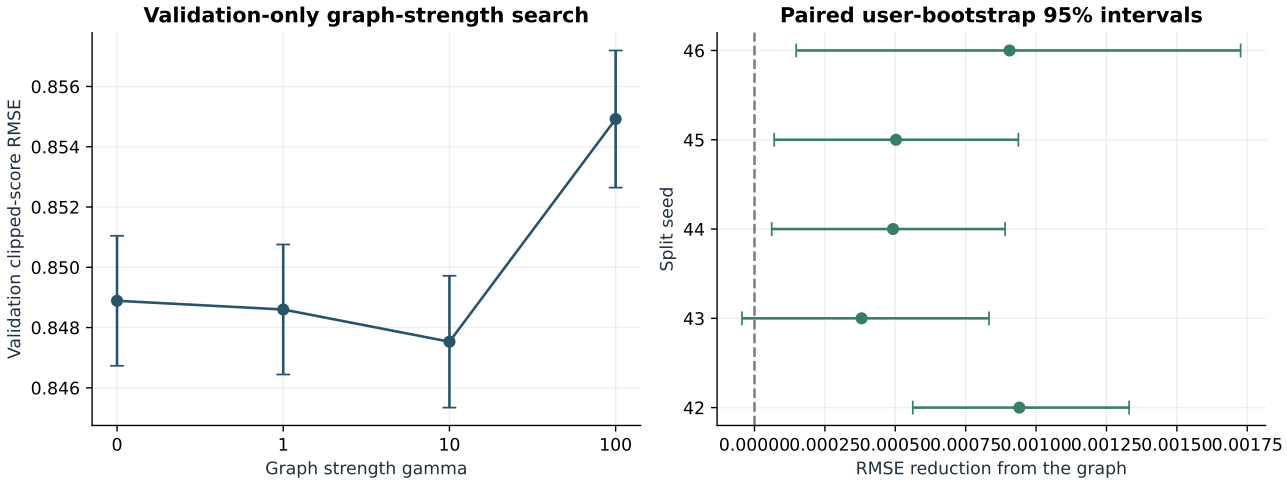


Figure 3: Validation graph-strength search and paired 95% intervals. Positive differences favour the graph. Strength 100 worsens validation error.

8.1 Robustness and Ablation

Condition	ADF	EP5	MF-MAP	GMRF-MAP
main	0.8980 ± 0.0049	0.8633 ± 0.0040	0.8507 ± 0.0028	0.8501 ± 0.0031
retain_0.25	0.9217 ± 0.0041	0.9199 ± 0.0051	0.9178 ± 0.0042	0.9100 ± 0.0038
retain_0.5	0.9102 ± 0.0039	0.8920 ± 0.0049	0.8810 ± 0.0039	0.8775 ± 0.0041
noise_0.1	0.9129 ± 0.0036	0.8910 ± 0.0036	0.8827 ± 0.0029	0.8797 ± 0.0030
noise_0.2	0.9386 ± 0.0030	0.9326 ± 0.0022	0.9317 ± 0.0023	0.9248 ± 0.0018
cold_items	1.0703 ± 0.0079	1.0781 ± 0.0068	1.0622 ± 0.0060	1.0545 ± 0.0077
cold_users	0.9837 ± 0.0137	0.9861 ± 0.0146	0.9935 ± 0.0115	0.9916 ± 0.0117

Table 3: Three-seed stress tests, mean ± SD. Cold-start rows use restricted test populations, so compare methods within each row.

We retain nested 25%/50% training subsets, or change 10%/20% of training labels to another star. Validation/test labels stay clean. Graphs are rebuilt from each condition; strength 10 is fixed, with validation still selecting epochs. The graph gain grows to 0.0077 RMSE at 25% retention and 0.0069 with 20% corruption. Smoothing mitigates some damage without restoring clean-data performance.

Holding out all training rows for 10% of movie IDs tests cold movies; genre links remain available. The graph improves 1.0622 to 1.0545 RMSE. For a corresponding held-out user test, ADF beats GMRF (0.9837 versus 0.9916): movie links cannot infer an unseen user's taste. On clean three-seed splits, rating-only edges work slightly better than mixed edges; genre-only edges worsen RMSE versus no graph. Top-5 improves on top-50. These are exploratory ablations, not test-driven changes to the main model.

9 Discussion

Repeated EP improves on ADF with the ordinal representation fixed. Later passes reconsider each observation after other beliefs have changed, at the cost of more memory and fitting time. This confirms a limited part of Matchbox's inference trade-off under fixed thresholds and static traits. It does not reproduce its online dynamics, personalized rating scales or published system-level results.

PreRec's transferable encoders and causal mechanisms target a different setting. Our experiments validate a narrower idea: sharing information through a fixed prior can help when evidence is thin. The matched $\gamma = 0$ ablation shows only a modest clean-data benefit; genre-only links and unseen users expose its limits. We therefore position the work as a reproducible study of approximate inference and graph regularization, not a new state-of-the-art recommender or a reproduction of PreRec.

10 Limitations and Future Work

Random-interaction splits do not represent future deployment. EP drops posterior covariance, while MAP treats traits and graph weights as fixed for prediction. Estimated edges can reflect exposure patterns rather than preference similarity. Artificial missingness and noise do not capture every real failure. Searches are finite, and confidence intervals hold training fixed. A useful next experiment would use chronological splits and compare posterior-aware graph inference with matched graph-free inference, controlling model capacity and calibration on validation data.

11 Conclusion

Repeated EP improves point and probability scores over ADF at a clear computational cost, without uniformly improving calibration. GMRF-MAP achieves 0.8499 RMSE, but most of its gain over EP is also present in graph-free MAP. Its isolated benefit is most useful with fewer or noisier training ratings. A movie graph can help an unseen movie when metadata is available; it cannot personalize a user with no observations.

References

[1] GroupLens Research. MovieLens 1M Dataset. `https://grouplens.org/datasets/movielens/1m/`. Official archive downloaded and SHA-256 verified on 4 November 2026.

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[3] Ziqian Lin, Hao Ding, Nghia Trong Hoang, Branislav Kveton, Anoop Deoras, and Hao Wang. Pre-trained recommender systems: A causal debiasing perspective. In *Proceedings of the 17th ACM International Conference on Web Search and Data Mining*, pages 424–433, 2024. doi:10.1145/3616855.3635779.

[4] David H. Stern, Ralf Herbrich, and Thore Graepel. Matchbox: Large scale online bayesian recommendations. In *Proceedings of the 18th International Conference on World Wide Web*, pages 111–120, 2009. doi:10.1145/1526709.1526725.

A Derivations and verification

Proper prior and movie update. For any nonzero x, $x^T Jx = \lambda\|x\|^2 + \gamma \sum_{\{i,j\}} w_{ij}(x_i - x_j)^2 > 0$. The conditional mean of movie trait Q_{ik} given all other movie values is $\gamma \sum_j w_{ij} Q_{jk}/(\lambda + \gamma D_{ii})$, and the conditional variance is $(\lambda + \gamma D_{ii})^{-1}$. For the MAP movie block $z = (Q_i, b_i)$, let $x_u = (P_u, 1)$ and $y_u = R_{ui} - \mu - b_u$. Setting the derivative to zero gives

$$\left[\sum_{u \in \mathcal{D}_i} x_u x_u^T + \operatorname{diag}\left((\lambda + \gamma D_{ii}) I_K, \lambda_b\right)\right] z = \sum_{u \in \mathcal{D}_i} x_u y_u + \gamma \begin{pmatrix} \sum_j w_{ij} Q_j \\ 0 \end{pmatrix}.$$

This system is positive definite. Its unique minimizer cannot increase the objective when the other blocks stay fixed.

Ordinal moment matching. For incoming score $\mathcal{N}(m, v)$ and observed star r, set $a = (c_{r-1} - m)/\sqrt{v}$, $b = (c_r - m)/\sqrt{v}$ and $Z = \Phi(b) - \Phi(a)$. With standard normal density ϕ, the tilted mean and variance are

$$m' = m + \sqrt{v}\frac{\phi(a) - \phi(b)}{Z}, \qquad v' = v\left[1 + \frac{a\phi(a) - b\phi(b)}{Z} - \left(\frac{\phi(a) - \phi(b)}{Z}\right)^2\right].$$

Subtract the incoming Gaussian natural parameters from $(1/v', m'/v')$ to obtain the outgoing message. This local calculation does not make the entire bilinear posterior exact.

Algorithmic validation. Nine implementation checks passed, including ADF–EP1 equality, site bookkeeping, reference-kernel equivalence and nonincreasing MAP energy. Independent 40/80-point quadrature agrees to 1.04×10^{-10} in the one-observation bilinear case. Hybrid inference underestimates the two reference variances by about 10% and omits a covariance of 0.0705. In a 12-variable Gaussian reference, the exact mean is $J^{-1}h$ and covariance J^{-1}. Gauss–Seidel mean RMSE falls from 0.138 at one sweep to 4.46×10^{-8} at 20 sweeps; Gibbs estimates variances as well, with sampling error. These are tractable diagnostics, separate from MovieLens prediction scores.

Verification. A fresh full-data run on seed 42 reproduced 146 recorded metric and selection values across 17 method rows within 10^{-8}, excluding timings. Split hashes matched. The recorded-results audit checked 229 comparison rows and the expected seed and condition coverage. See `Code/README.md` for the exact commands, dependencies and seeds, and `Results/` for raw tables, uncertainty intervals, logs and figures.

Latent dimension. A separate three-seed EP5 experiment obtained RMSE of 0.8760 ± 0.0059 with two traits and 0.8633 ± 0.0040 with eight. These runs are separate from the five-split benchmark.

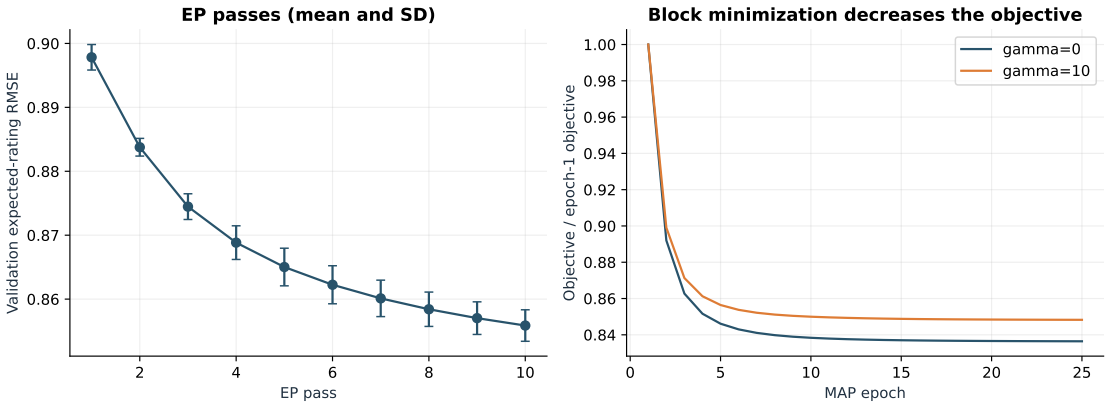


Figure 4: Learning traces: validation RMSE over EP passes and the normalized MAP objective over epochs, averaged across the recorded splits.